Last Time

Two Independent Samples
  Sampling Distribution of $\overline{X}_1 - \overline{X}_2$

Behrens-Fisher Problem

Dependent Samples

Advantages to dependent sample designs

Today

Dependent Samples

Advantages to dependent sample designs

Effect Sizes for Mean Comparisons

- Cohen's d
- Hedges' g
- Glass' Δ

Power

- Factors affecting power

Alzheimer's Diagnostic

Finger tapping speed can be an indication of a number of neurological disorders. Christianson and Leathem (2004) developed a computerized diagnostic test that, for healthy individuals, has a mean of $\mu = 59$ ~~and a standard deviation of $\sigma = 7$~~.

A sample of $N = 10$ Alzheimer's patients takes the test, and we observe their mean $\overline{X} = 48.90$ and standard deviation $s = 13.36$.

Difference Scores

Patient	1	2	3	4	5	6	7	8	9	10
Before Treatment	36	46	76	58	49	35	46	51	60	32
After Treatment	42	46	75	65	55	33	49	53	54	40
Difference	-6	0	1	-7	-6	2	-3	-2	6	-8

Before	After
$\overline{X} = 48.90$	$\overline{Y} = 51.20$

$$\overline{D} = -2.3$$
$$s_D = 4.55$$
$$N = 10$$

In R

```
> before <- c(36,46,76,58,49,35,46,51,60,32)
> after <- c(42,46,75,65,55,33,49,53,54,40)
> t.test(x=after,y=before,mu=0,paired=TRUE,alternative='less')

 Paired t-test

data:  before and after
t = -1.5995, df = 9, p-value = 0.07209
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf 0.3359742
sample estimates:
mean difference
           -2.3
```

Designs with dependent samples are *good*

They are *more powerful*, which means we can reject false H_0s more easily.
The reason for this has to do with $\sigma_{\overline{X}-\overline{Y}}$, the standard error of the mean difference.

$$\sigma_{\overline{X}-\overline{Y}} = \sqrt{\sigma^2_{\overline{X}} + \sigma^2_{\overline{Y}} - 2\rho\sigma_{\overline{X}}\sigma_{\overline{Y}}}$$

If $\rho > 0$, the standard error is reduced.

- Typically, in matched samples or repeated measures, $\rho > 0$

Variance Comparisons

$$
\begin{aligned}
s^2_{\text{before}} &= 178.54 \\
s^2_{\text{after}} &= 150.62 \\
r &= 0.94
\end{aligned}
$$

$$
\begin{aligned}
s^2_P &= 164.58 \quad (df = 18) \\
s^2_{\text{before}} + s^2_{\text{after}} &= 329.16
\end{aligned}
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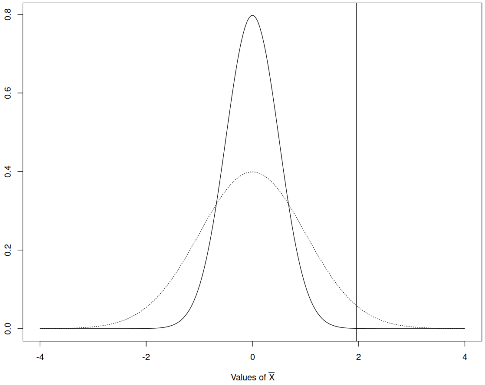
s^2_{\text{before}} + s^2_{\text{after}} - 2\, r\, s_{\text{before}}\, s_{\text{after}} &= 329.16 - 2(0.92)(13.36)(12.28) = 20.68 \\
s^2_D &= 20.68 \quad (df = 9)
\end{aligned}
$$

Variance Comparisons

```
> var(before)
[1] 178.5444

> var(after)
[1] 150.6222

> cor(before,after)
[1] 0.9405715

> var(before) + var(after) - 2*cor(before,after)*sd(before)*sd(after)
[1] 20.67778

> var(before-after)
[1] 20.67778
```

Advantages to Dependent Samples Designs

- Control extraneous sources of variation (nuisance or confounding variables)
- *Smaller standard error



Disdvantages to Dependent Samples Designs

- Order effects

  Alzheimer's disease *progresses*.
  - Two groups: one group gets the drug first and is then taken off the regimen for retest, and the second group is placed on the regimen after the first test.
  - While the drug gets time to have an effect, the disease gets worse.
  - Some questions require controlling for order of treatments
- Matched pairs are not randomly assigned to conditions

These disadvantages can usually be overcome with good experimental design.

Does Money Buy Happiness?

```
> t.test(money~happy,alternative="less",var.equal=TRUE)

 Two Sample t-test

data:  money by happy
t = -2.3473, df = 871, p-value = 0.009566
alternative hypothesis: true difference in means between group
FALSE and group TRUE is less than 0
95 percent confidence interval:
     -Inf -1768.53
sample estimates:
mean in group FALSE  mean in group TRUE
           17924.87            23849.19
```

Is a Significant Effect Important?

$$z_{\overline{X}} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{N}}$$

$$t_{\overline{X}} = \frac{\overline{X} - \mu_0}{s/\sqrt{N}}$$

Is a Significant Effect Important?

$$z_{\overline{X}} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{N}} \qquad t_{\overline{X}} = \frac{\overline{X} - \mu_0}{s/\sqrt{N}}$$

$5,924.32 in 1980 is $22,605.77 today.

Effect Size

Definition

A standardized measure that quantifies the strength or magnitude of a phenomenon or relationship between variables in a statistical test in a way that is independent of sample size.

Why?

There is a distinction between a significant effect and a meaningful one.

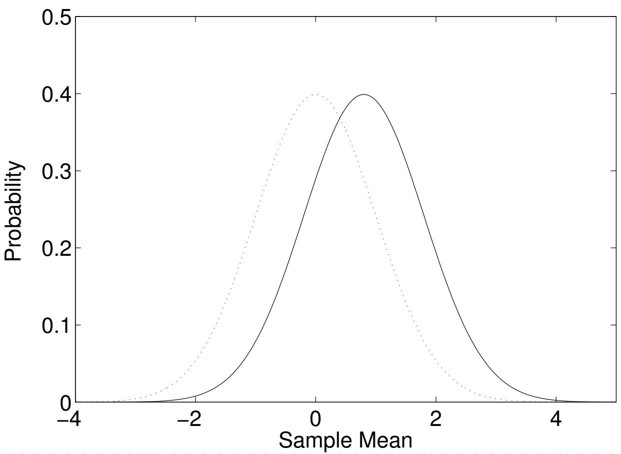
- Because an effect size is independent of sample size we can talk about meaningfulness when a difference may be statistically significant due to sample size alone.
- Meta-analysis: standardizing the size of differences allows allows comparisons of results across different studies regardless of sample size or study design.
- Power: knowing the expected effect size helps researchers design studies with appropriate sample sizes to detect meaningful effects.

Effect Size Measures for Mean Comparisons

d-measures (standardized differences)

- Cohen's d
- Hedges' g
- Glass' Δ

Effect Size



Cohen's d

Z-test

$$d = \left| \frac{\overline{X} - \mu_0}{\sigma} \right|$$

T-test

$$d = \left| \frac{\overline{X} - \mu_0}{s} \right|$$

Independent Samples T-test

$$d = \left| \frac{\overline{X}_1 - \overline{X}_2}{s_P} \right|$$

Cohen's Convention

d	Size
.2	Small
.5	Medium
.8	Large

GSS Example I

```
> library('effectsize')
> t.test(money,mu=12513)

 One Sample t-test

data:  money
t = 13.031, df = 872, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 12513
95 percent confidence interval:
 21548.13 24752.30
sample estimates:
mean of x
 23150.21

> cohens_d(money,mu=12513)
Cohen's d |       95% CI
------------------------
0.44      | [0.37, 0.51]

- Deviation from a difference of 12513.
```

GSS Example II

```
> t.test(money~happy,alternative="less")

 Welch Two Sample t-test

data:  money by happy
t = -3.0283, df = 161.28, p-value = 0.001432
alternative hypothesis: true difference in means between group 0 and group 1 is less than 0
95 percent confidence interval:
      -Inf -2687.825
sample estimates:
mean in group 0 mean in group 1
       17924.87        23849.19

 > cohens_d(money~happy,alternative="less",pooled_sd=TRUE)
Cohen's d |         95% CI
-------------------------
-0.25     | [-Inf, -0.07]

- Estimated using pooled SD.
- One-sided CIs: lower bound fixed at [-Inf].
```

Hedges' g

Adjusts for smaller samples:

$$J(df) = \frac{\Gamma(df/2)}{\sqrt{df/2}\,\Gamma((df-1)/2)}$$

$$g = dJ(df)$$

```
> hedges_g(money~happy,sd_pooled=TRUE)

Hedges' g |         95% CI
--------------------------
-0.25     | [-0.45, -0.04]

- Estimated using pooled SD.
```

Glass' Δ

Let "group 2" be a control group with "more stable" variance. Then

$$\Delta = \frac{\overline{X}_1 - \overline{X}_2}{s_2}.$$

If "happy" is group 2:

```
> glass_delta(money~happy)

Glass' delta (adj.) |         95% CI
------------------------------------
-0.24               | [-0.39, -0.08]
```

Power

State of the World	Decision	
	H_0 True	H_0 False
H_0 True	1 - α	α
H_0 False	β	1-β

Power

State of the World	Decision H_0 True	H_0 False
H_0 True	1 - α	α
H_0 False	β	1-β

Define significance level:

$$\alpha = Pr(\text{Type } I \text{ error}) = Pr(\text{Reject } H_0 | \, H_0 \text{ true})$$

Power

State of the World	Decision H_0 True	H_0 False
H_0 True	1 - α	α
H_0 False	β	1-β

Define significance level:

$$\alpha = Pr(\text{Type } I \text{ error}) = Pr(\text{Reject } H_0 | \, H_0 \text{ true})$$

Define power:

$$\begin{aligned} 1 - \beta &= 1 - Pr(\text{Type } II \text{ error}) \\ &= Pr(\text{Reject } H_0 \,|\, H_0 \text{ false}) \end{aligned}$$

Error Probabilities

- The probability of rejecting a true H_0 is α. We set this probability directly when we select a significance level.
- The probability of rejecting a false H_0 depends on the effect size! If we specify an effect size, we can compute this probability, which is called *power* and equals $1 - \beta$

Error Probabilities

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- The probability of rejecting a false H_0 depends on the effect size! If we specify an effect size, we can compute this probability, which is called *power* and equals $1-\beta$
  Power is the ability of a test to detect/reject a false null hypothesis.
  We can only compute $1-\beta$ if we know H_1 - if we specify an effect size.
  Typically we don't know H_1

Power and Significance

- Significance levels α are selected in advance, based on practical considerations.
- Issues of power are often not considered until after a failure to reject H_0. However, there are ways to insure good power, which should be used *before* testing H_0.

Factors Affecting Power

1. Effect Size
2. Sample Size
3. Significance level
   - One vs. two-tailed tests